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3 December 1953
SUPERSEDING
PA-PD-16 (Rev 1)
7 August 1953

MILITARY SPECIFICATION

CARTRIDGES, HE: T50E2 FOR 75-MM GUN, T83E1;
HE, T64 FOR 76-MM TANK GUN, T91
ASSEMBLING, MARKING, AND PACKING

1. SCOPE

1.1 This specification covers assembling, marking, and packing of high explosive cartridges for use in 75-mm and 76-mm guns.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, drawings, and publications, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition

DRAWINGS

ORDNANCE CORPS

75-1-292 -Cartridge, HE, T50E2, W/Supplementary Charge, W/O Fuze for 75MM Gun, T83E1 Assembly and Marking Diagram

75-1-293 -Cartridges, HE and TP, T64 W/Fuzes for 76MM Tank Gun, T91. Assembly and Marking Diagram

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M11-1-Complete Rounds, Fixed, HE For 57MM, 75MM and 105MM Rifles, and 75MM, 76MM and 90MM Guns

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(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 The cartridges shall comply with all requirements specified on Drawing 75-1-292, or Drawing 75-1-293, as applicable, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Propelling charge.-The weight of the propelling charge shall not differ by more than 0.05 ounce avoirdupois from that designated by the loading authorization for the particular lot of powder used. The contractor shall weigh and check-weigh each propelling charge. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that used to make the first weighing.

3.4 Crimp.

3.4.1 T50E2 cartridge.-The cartridge case shall withstand a pull of 5000 pounds minimum without separating from the shell. Cartridges using T6E3B1 steel cartridge cases shall withstand a pull of 5000 pounds, minimum, to 11,000 pounds, maximum, without separating from the shell when tested as specified in 4.4.

3.4.2 T64 cartridge.-The cartridge case shall withstand a pull of 5000 pounds minimum, to 11,000 pounds maximum, without separating from the shell when tested as specified in 4.4.

3.5 Proving ground requirements for the cartridges.

3.5.1 Average velocity.-The difference between the velocities of any two cartridges fired in the velocity test shall not exceed 3 percent of the mean velocity of all the cartridges fired in this phase of the test. The average velocity shall not differ from the specified velocity (see 6.2) by more than 2 percent after reducing results to standard conditions when tested as specified in 4.5.1.1.

3.5.2 Premature bursts.-There shall be no premature bursts or evidence thereof when tested as specified in 4.5.1.2. A "premature burst" shall be defined as a functioning of the cartridge in the gun bore or in flight before impact.

3.5.3 Safety.-The cartridge shall function satisfactorily when assembled with powder charges adjusted to give pressures of 112 percent of the average pressure given by the propelling charge loaded in the particular ammunition lot when tested as specified in 4.5.1.2.

3.5.4 Shell functioning.-Not more than one shell shall fail to function satisfactorily when tested as specified in 4.5.1.3.

3.6 Repainting.-The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated.

3.7 Workmanship.

3.7.1 All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.7.2 Care shall be exercised at all times so that the rotating band of each shell is not damaged by being nicked or burred.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.-A lot shall consist of all cartridges assembled by one manufacturer under one contract in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain:

a. steel cartridge cases of new manufacture made by one manufacturer in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

b. loaded shell of the same weight zone, made by one manufacturer at one plant, in accordance with same drawing, same drawing revision, same specification, and same specification revision using consecutive lots insofar as practicable.

c. loaded fuzes from not more than two lots.

d. supplementary charges made in accordance with the same drawing, same drawing revision, same specification and same specification revision.

e. loaded primers from not more than two lots. These lots should be consecutive lots, if possible (not a requirement).

f. propellant from not more than one lot, or of not more than two lots for dualgran charge.

g. cartridges of one type only.

4.2 Sampling.-Number of samples, acceptance and rejection criteria used for determination of lot acceptance for test as specified in 4.4 shall be in accordance with Standard Inspection Procedure ORD-SIP-M11-1. Twenty cartridges shall be selected from each lot by the inspector for proving ground test as specified in 4.5.

4.3 Inspection.

4.3.1 Inspection shall be as specified in Specification MIL-G-2550.

4.3.2 Gaging.-After the cartridge has been assembled, the contractor shall gage each round of ammunition in a profile and alignment gage (commonly known as a chamber gage) pertaining to this round. If the cartridge fails to enter the profile and alignment gage freely, it shall be rejected and subject to correction and retest. Drawings of Government profile and alignment gage will be furnished to the contractor for his information upon application to the inspector.

4.4. Test of crimp.-Sample cartridges shall be tested in a tensile testing machine satisfactory to the inspector, to determine the security of the crimp of the cartridge case to the shell, by applying a load in the axial direction, rate of travel of testing machine head to be .100 to .125 inch per minute inclusive to determine compliance with 3.4.

4.5 Proving ground tests.

4.5.1 Procedure.

4.5.1.1 Velocity.

4.5.1.1.1 Cartridges issued with live fuzes.-Five of the sample cartridges shall be fired from a weapon for which the cartridge is standard for impact on firm ground using M51 Fuzes set "superquick" action. The velocities shall be measured and observation shall be made for functioning to determine compliance with 3.5.1.

NOTE: The firing shall be conducted using non-impact type velocity screens.

4.5.1.1.2 Cartridges issued with closing plugs.-Five of the sample cartridges shall be fired for impact on firm ground after removing closing plug, washer, and spacer and assembling with any lot of accepted fuzes set "superquick". Velocities shall be measured and observation shall be made for functioning to determine compliance with 3.5.1.

NOTE: The firings shall be conducted using non-impact type velocity screens.

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4.5.1.2 Safety.

4.5.1.2.1 Cartridges issued with live fuzes.-Five of the sample cartridges shall be disassembled and reassembled with powder charges adjusted to give pressure of 112 percent of the average pressure given by the propelling charge loaded in the particular ammunition lot. This value of pressure appears on the data cards of each ammunition lot. The cartridges shall be fired for impact on firm ground (with M51 Fuzes set "superquick") and observation shall be made for premature bursts and order of functioning to determine compliance with 3.5.3.

4.5.1.2.2 Cartridges issued with closing plugs.-Five of the sample cartridges shall be disassembled and reassembled with powder charges adjusted to give pressures of 112 percent of the average pressure given by the propelling charge loaded in the particular ammunition lot. The cartridges shall be fired after removing closing plug, washer, and spacer and assembling with M51 PD Fuzes. Observation shall be made for premature bursts and order of functioning to determine compliance with 3.5.3.

4.5.1.3 Functioning.

4.5.1.3.1 Cartridges issued with fuzes.-Ten of the sample cartridges shall be fired for impact on firm ground, in a weapon for which the cartridge is standard. When M51A5 PD Fuzes are used, set for "superquick" action. Observation shall be made for order of functioning to determine compliance with 3.5.4.

4.5.1.3.2 Cartridges issued with closing plugs.-The closing plug, washer, and spacer, shall be removed from ten sample cartridges and assembled with any lot of accepted fuzes. The sample cartridges shall be fired for impact on firm ground from a weapon for which the cartridge is standard. When M51A5 fuzes are used, set for "superquick" action. Observation shall be made for order of functioning to determine compliance with 3.5.4.

NOTE: At the option of the proof officer, the cartridge may be tested using either M51 series PD fuzes assembled with inert delay plunger assemblies; M57 PD fuzes assembled with M21A4 boosters; or T177 PD fuzes.)

4.5.2 Acceptance.-The lot shall be considered acceptable provided the sample cartridges fired and complied with 3.5.1, 3.5.2, 3.5.3 and 3.5.4. The lot shall be considered eligible for retest under the provisions of 4.5.3 provided no premature burst occurred and if:

a. the difference between the velocities of any two cartridges fired in the velocity test exceeds 3 percent but does not exceed 4 percent of the mean velocity of all rounds fired in this phase of test, or

b. the average velocity differs from the specified velocity by more then 2 percent, but not more than 3 percent, or

c. not more than two cartridges fail to function satisfactorily.

The lot shall be rejected if any premature burst occurs, or if in the phase being tested, the limits for eligibility for retest are exceeded.

4.5.3 Retest.-A retest as provided for in 4.5.2 may be made if requested by the contractor using double the number of samples used in the original test. The lot shall be considered acceptable on retest provided no premature burst occured and provided that in the phase being tested, (i.e., velocity or functioning) the rounds fired and complied with the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.5.4.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking.-Packaging, packing, labeling and marking shall be as specified on the applicable drawing.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with proving ground tests of the cartridge to save expense upon agreement between the United States and the contractor to that effect. In case the cartridge specification does not cover all of the proving ground tests specified for the components, the additional tests specified in the component specifications should be conducted.

6.2 Information concerning the specified velocity may be obtained from the procuring agency.

NOTICE.-This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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